

Low-Rank Functional Approximation via Chebyshev–Tucker Decomposition

1 Problem Setting

Let

$$f : [-1, 1]^3 \rightarrow \mathbb{R}$$

be a smooth function. We seek a compact approximation of f that avoids the cubic cost of full tensor discretization while preserving high accuracy.

2 Chebyshev Tensor Expansion

Using Chebyshev polynomials T_i , a truncated tensor-product expansion of f is

$$f(x, y, z) \approx \sum_{i=0}^{d_x} \sum_{j=0}^{d_y} \sum_{k=0}^{d_z} C_{ijk} T_i(x) T_j(y) T_k(z), \quad (1)$$

where $C \in \mathbb{R}^{(d_x+1) \times (d_y+1) \times (d_z+1)}$ is the Chebyshev coefficient tensor. Direct construction of C is expensive for large degrees.

3 Tucker Approximation

We approximate the coefficient tensor using a Tucker decomposition

$$C_{ijk} \approx \sum_{a=1}^{R_x} \sum_{b=1}^{R_y} \sum_{c=1}^{R_z} A_x(i, a) A_y(j, b) A_z(k, c) G_{abc}, \quad (2)$$

where A_x, A_y, A_z are factor matrices and G is a small core tensor. This reduces storage from cubic to linear in each dimension.

4 Value-Space Formulation

Let $\{x_i\}, \{y_j\}, \{z_k\}$ be training nodes (typically Chebyshev points), and define Vandermonde matrices

$$T_x(i, p) = T_p(x_i), \quad T_y(j, q) = T_q(y_j), \quad T_z(k, r) = T_r(z_k).$$

Define projected bases

$$U_x = T_x A_x, \quad U_y = T_y A_y, \quad U_z = T_z A_z.$$

The reconstructed values satisfy

$$\hat{F}_{ijk} = \sum_{a,b,c} U_x(i, a) U_y(j, b) U_z(k, c) G_{abc}. \quad (3)$$

5 Alternating Least Squares

The unknowns (A_x, A_y, A_z, G) are computed by minimizing

$$\min_{A_x, A_y, A_z, G} \|F - \hat{F}\|_F^2$$

using Alternating Least Squares (ALS).

Mode-1 Update

Fix A_y, A_z and define

$$U_y = T_y A_y, \quad U_z = T_z A_z.$$

Let

$$Z_{yz} = U_z \otimes U_y, \quad F_{(1)} \in \mathbb{R}^{N_x \times (N_y N_z)}$$

denote the mode-1 unfolding of F .

Solve

$$\min_{\tilde{G}_x} \left\| \text{vec}(F_{(1)}) - (Z_{yz} \otimes T_x) \text{vec}(\tilde{G}_x) \right\|_2^2,$$

then compute the truncated SVD

$$\tilde{G}_x \approx U_x^{(c)} \Sigma_x V_x^\top.$$

Set

$$A_x = U_x^{(c)}(:, 1:R_x), \quad G_{(1)} = \Sigma_x V_x^\top.$$

Mode-2 and Mode-3 Updates

The updates for A_y and A_z are obtained analogously by cyclic permutation of tensor modes, using unfoldings $F_{(2)}$ and $F_{(3)}$.

6 Metric-Aware Orthonormalization

To maintain numerical stability, factor matrices are orthonormalized with respect to

$$G_x = T_x^\top T_x, \quad G_y = T_y^\top T_y, \quad G_z = T_z^\top T_z,$$

so that

$$A_x^\top G_x A_x = I, \quad A_y^\top G_y A_y = I, \quad A_z^\top G_z A_z = I.$$

7 Algorithm

Algorithm 1 Chebyshev–Tucker ALS

- 1: Choose degrees (d_x, d_y, d_z) and ranks (R_x, R_y, R_z)
 - 2: Sample f on a tensor-product grid to obtain F
 - 3: Initialize A_x, A_y, A_z, G
 - 4: **for** iter = 1, ..., maxIter **do**
 - 5: Update (A_x, G) via mode-1 least squares
 - 6: Update (A_y, G) via mode-2 least squares
 - 7: Update (A_z, G) via mode-3 least squares
 - 8: **end for**
 - 9: Return (A_x, A_y, A_z, G)
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8 Final Approximation

The resulting approximation is

$$\hat{f}(x, y, z) = \sum_{a,b,c} \left(\sum_i A_x(i, a) T_i(x) \right) \left(\sum_j A_y(j, b) T_j(y) \right) \left(\sum_k A_z(k, c) T_k(z) \right) G_{abc},$$

which can be evaluated efficiently at arbitrary points.

9 Conclusion

This approach combines spectral accuracy from Chebyshev polynomials with low-rank Tucker structure. The ALS algorithm yields a stable and scalable method for multivariate functional approximation with significantly reduced complexity.